



OSA Data Assessment Report

Personal details					Weight & Height:	93 kg (205 lbs) 182 cm (5'11.7")
	Gender:	M	Age:	58	BMI:	28.1

Clinical background: Hyperlipidaemia, Psoriasis, Smoker, OSA, CPAP intolerance (was treated with nasal mask since 2013)

Patient presentation: Reports of unrefreshing sleep (chronic fatigue and desire to sleep), difficulty falling back asleep after early-morning awakenings, and loud snoring occurring almost every night (sometimes more, sometimes less). Bed partner occasionally notices pauses in breathing during sleep. Excessive daytime sleepiness is present (tiredness and sleepiness even after exercise). Additional symptoms include waking up with dry mouth or thirst, frequent nighttime urination, jaw pain and clicking (pain in the left TMJ with clicking upon opening), nighttime teeth grinding, morning headaches, and occasional tinnitus ("sometimes"). Tobacco use and overweight status reported. Alcohol consumption is approximately one bottle of beer per week.

Patient goals: fatigue, enhanced sleep quality, increased daily energy levels, and reduced health risks.

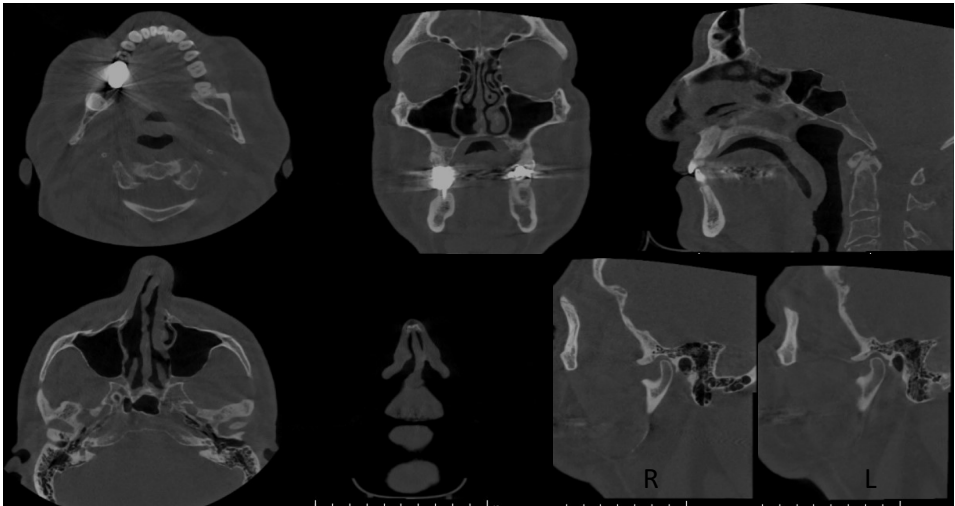
Sleep Study Data

Sleep duration	8h and 16min
Sleep efficiency	93.5%
AHI	59.4 (Severe OSA)
ODI	30.1 (401 decrease of oxygen)
O₂ Nadir	77%
Snoring	avg 42 dB, max 68 dB

Observations:

- Data indicate severe OSA, with an AHI of 59.4 over a total sleep time of 8 hours and 16 minutes, and high sleep efficiency of 93.5%.
- Desaturation events were recorded with an ODI of 30.1, including 401 instances of oxygen drop and a minimum oxygen saturation (O₂ nadir) of 77%.
- Snoring was present throughout the night, with an average intensity of 42 dB and a peak of 68 dB.

Structural Observations from Imaging Data



Key Observations	Details
Obstruction Sites	Primary obstruction at the velopharyngeal level.
Soft Palate & Uvula	Elongated and swollen, with the uvula extending inferiorly toward the epiglottis and partially obstructing the oropharynx.
Tongue Position	Posteriorly positioned tongue base encroaching the airway; possible reduced muscular tone may be present.
Bite & Jaw Structure	Increased overjet, excessive overbite, high-vaulted palate, and narrow arches.
Hyoid Bone	Not clearly visualized but is inferred to be positioned significantly inferior to the mandibular plane.
Nose & Sinuses	Post-septoplasty changes with septal irregularities and possible nasal valve collapse. The right inferior turbinate appears absent. Soft tissue density noted in the right maxillary sinus.
TMJ	Right condylar head is posteriorly positioned. Left condylar head is slightly flattened and posteriorly positioned.

Conclusion: The provided imaging data suggests potential airway obstruction primarily related to velopharyngeal narrowing, soft palate elongation, posterior tongue base positioning, and a low hyoid bone position. Craniofacial structural features may contribute to pharyngeal space limitation. Posterior condylar positioning, with mild flattening on the left, may require ongoing monitoring.

Possible Treatment Considerations

- Given the severity of OSA and intolerance to CPAP, a combined treatment strategy may be considered, integrating oral appliance therapy with adjunctive options such as neuromuscular stimulation or surgical intervention.
- In light of the patient’s anatomical findings and history of CPAP intolerance, oral appliance therapy may offer benefit, though should be approached with clinical caution.
- Temporomandibular joint symptoms and radiographic findings suggest the need for monitoring if mandibular advancement is pursued.
- Vertical opening and anterior repositioning may assist in improving pharyngeal patency by addressing tongue base and velopharyngeal collapse.

Device Design Data Considerations

Parameter	Data-Based Consideration
Mandibular Advancement	2mm from edge-to-edge positioning
Pre-set	+1mm
Vertical Opening	5mm anterior
Anterior Window	Large, allowing tongue access to upper incisors
Retention Features	Ball clasps for future adjustments, elastic hooks for stabilization
Material	High-density acrylic (hygienic, adjustable)
Anterior Acrylic	Full coverage
Coverage	Full coverage
Clinical Notes	Consider using elastic bands, nose cones, and nasal strips from the start to support nasal breathing. Consider selecting a device that enables substantial mandibular advancement over time.

Recommendations for Further Evaluation

- A follow-up ENT evaluation may be considered to reassess nasal anatomy post-septoplasty, including valve collapse and turbinate status, and to assess candidacy for possible surgical intervention.
- Nasal airflow optimization—using nose cones, nasal strips, or surgery—may enhance treatment effectiveness and overall breathing quality.
- Encourage positional therapy to minimize supine sleep time and support airway stability.
- Consider lifestyle modifications, including weight reduction and improved sleep hygiene, as supportive strategies for symptom improvement.
- Neuromuscular stimulation therapy (e.g., eXciteOSA) may be considered as a complementary approach to support pharyngeal muscle tone and reduce collapsibility.

- TMJ symptoms and condylar morphology may require ongoing monitoring if oral appliance therapy is initiated.
- Consider repeating a sleep study in three months while using the oral appliance to evaluate treatment effectiveness.
- If symptoms persist despite therapy, consider a follow-up DISE with the appliance in place to assess upper airway dynamics and guide further treatment adjustments.

Oral Appliance Options for Consideration

Device		Key Features
Emerald Herbst		Strong, lightweight, durable, high-density acrylic for hygiene
Respire Herbst Pink AT		Promotes mandible stabilization, high-density acrylic, metal mesh embedded
Daynaflex Herbst		Accurate fit, enhanced tongue space, stain-resistant PMMA

Disclaimer

This AI-generated report is intended as a data assessment tool to support clinical decision-making. It does not constitute a medical diagnosis, treatment recommendation, or radiological interpretation. All imaging findings, including CBCT references, should be reviewed and confirmed by a certified radiologist or qualified healthcare provider. Final treatment decisions must be made by licensed professionals based on a comprehensive clinical evaluation. No liability is assumed for medical decisions made solely on the basis of this report.